

1-Bit Half Subtractor Using Dataflow Modeling

Aim

To design and verify a 1-bit half subtractor in Verilog using dataflow modeling.

Code

```
module half_subtractor(  
    input A,  
    input B,  
    output Difference,  
    output Borrow  
);  
  
    assign Difference = A ^ B;  
    assign Borrow = (~A) & B;  
  
endmodule
```

Truth Table / Observation

A	B	Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Observation: All 4 input combinations were simulated and matched the expected half-subtractor behavior. The simulation generated `half_subtractor.vcd` and completed at 40 ns.

Conclusion

The 1-bit half subtractor was successfully implemented using Verilog dataflow modeling. XOR generates the difference, while the inverted minuend combined with the subtrahend generates the borrow.